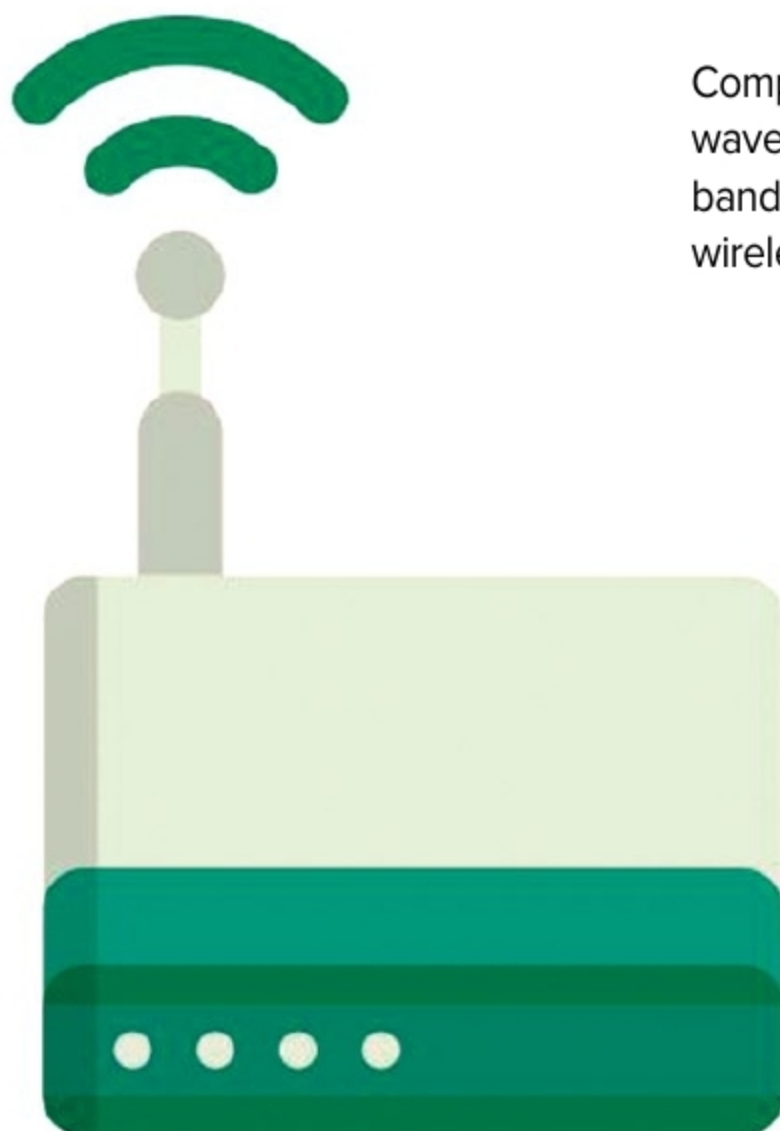


Wi-Fi In 60GHz BAND To Solve Bandwidth Challenges



Compared to 2.4GHz and 5GHz bands, millimetre-wave-based 60GHz Wi-Fi offers a low-interference, bandwidth-rich spectrum that delivers increased wireless capacity and ultra-low latency

Wi-Fi is now also being used in 60GHz band, which is a millimetre-wave unlicensed band. The millimetre-wave spectrum is much in demand for providing speeds in gigabits per seconds. Millimetre waves are used to provide high data rates using advanced technologies like massive MIMO and the like.

However, there are some drawbacks of millimetre-wave. It has fast attenuation at a distance, and it is likely to be absorbed by the atmosphere. Therefore it degrades fast with distance and can be used for shorter distance propagation only.

Scopes of the 60GHz band

At 60GHz, radio waves propagate in the line-of-sight for better transmission and reception. Due to high data rate advantage of the 60GHz spectrum band, Wi-Fi standards are being developed to achieve data rate of 10Gbps. This data rate is beneficial to create a backbone network using multiple line-of-sight access points in mesh connectivity.

Terragraph is one such network, backed by Facebook. It will provide a city-wide backbone network for faster data connectivity in an ubiquitous way. It is also being seen as a replacement for fibre and a cheaper way of backhauling data. "With Terragraph, our goal is to enable people living in urban areas to access high-quality connectivity that can help create new opportunities and

Wi-Fi is a widely-used technology for wireless Internet connectivity. It uses the unlicensed spectrum for radio waves, mostly in the 2.4GHz - 5GHz band. Wi-Fi has been coping with other wireless access technologies through its rapid advancements. New standards, developed by IEEE, have kept it a relevant technology with time. It has provided necessary data rates for keeping up with next-generation applications. There have also been significant improvements in reliability and security through WPA2 and WPA3 standards.



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